Akoizo Original Practice - Form 1 - Answer key

Division B | Botany | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) NPP equals…

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

A. Gross production minus producer respiration

B. Gross production plus all respiration

C. Only nighttime growth

D. Total ecosystem carbon storage

Correct: A

2. (2 points) Higher water potential: -0.4 or -0.8 MPa?

A. -0.4 MPa

B. -0.8 MPa

C. They are identical

D. Neither can be compared

Correct: A

3. (2 points) What does the plateau suggest?

Unit: Photosynthesis and plant energy.
Worked-case data: Low light: 5 bubbles per minute. Medium light: 10 bubbles per minute. High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.
Separate investigation observations:
Low light: 5 bubbles per minute.
Medium light: 10 bubbles per minute.
High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.

A. Each bubble must have exactly the same volume

B. Light never affects photosynthesis

C. Plants do not respire

D. Another factor may limit the rate at high light

Correct: D

4. (2 points) Hydroponic roots still need…

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

A. Oxygen and suitable nutrient conditions

B. No oxygen

C. Only pure water forever

D. No selective membranes

Correct: A

5. (2 points) Guard cells regulate…

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

A. Root hair length only

B. Stomatal opening

C. Every xylem vessel directly

D. Pollen chromosome number

Correct: B

6. (2 points) The example most strongly supports…

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

A. Every gymnosperm

B. Monocot

C. Moss

D. Fern

Correct: B

7. (2 points) Do all plants directly fix atmospheric nitrogen?

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

A. Only every woody stem

B. No

C. Yes

D. Only every green leaf

Correct: B

8. (2 points) Phloem flow is best described as…

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

A. Always upward

B. Only from roots to leaves

C. Source to sink

D. Always downward

Correct: C

9. (2 points) Expected recessive phenotype in Aa x Aa with complete dominance?

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

A. 3/4

B. All offspring

C. 1/4

D. 1/2

Correct: C

10. (2 points) Similar traits can arise independently through…

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

A. A display stretch

B. A universal ladder

C. Convergence

D. Only identical ancestry

Correct: C

11. (4 points) Calculate the example NPP.

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

answer: 100 - 40 = 60 carbon units per stated interval. (4 points).

12. (4 points) Predict movement from -0.2 to -0.5 MPa.

answer: Toward -0.5 MPa along an available pathway, from higher to lower potential. (4 points).

13. (6 points) Solve this case and explain your reasoning: Low light: 5 bubbles per minute. Medium light: 10 bubbles per minute. High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.

reasoning: Compare low and medium light: the indicator increases.
Compare medium and high light: the indicator stays constant.
Test another limiting factor while keeping the method consistent. (3 points).

conclusion: Another factor may limit the rate at high light (3 points).

14. (4 points) Distinguish hydroponics and aquaculture.

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

answer: Hydroponics is a plant cultivation method using nutrient solution; aquaculture cultivates aquatic organisms and can include different systems. (4 points).

15. (4 points) Explain the stomatal tradeoff.

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

answer: Closing pores conserves water but can restrict CO_2 entry and hence photosynthesis. (4 points).

16. (4 points) Why avoid one-trait identification?

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

answer: Traits have exceptions and can converge; combined reproductive and anatomical evidence is more reliable. (4 points).

17. (4 points) Why measure water use and yield together?

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

answer: They reveal efficiency and tradeoffs that one growth measure can hide. (4 points).

18. (4 points) Why use multiple anatomical traits for classification?

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

answer: Single traits can vary or evolve convergently; combined evidence supports a more reliable identification. (4 points).